Is Network Reconfiguration Worth It at Infrastructure Renewal? A Feasibility-Gated Economic Screening of Low-Voltage Distribution Re-Layout during Undergrounding

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Abstract

Scheduled renewal of urban infrastructure, such as the conversion of overhead electricity distribution lines to underground cable, commits civil works that could also be used to reconfigure the network. Whether such reconfiguration is worth undertaking is rarely evaluated directly: optimization studies usually compare a redesigned network with a greenfield or forced alternative rather than with retention of the existing layout, and the existing layout is often only partially recorded. This study develops a feasibility-gated screening method that estimates the economic value of reconfiguration at renewal while keeping like-for-like replacement as an admissible option. Incumbent low-voltage networks were reconstructed as ensembles from open municipal records and OpenStreetMap data in three Japanese study zones with different levels of record observability. Candidate redesigns were generated as weighted Steiner trees on the road network and admitted only if they satisfied radiality, coincident-peak ampacity, transformer-capacity, and voltage-drop limits; their life-cycle net present cost was then compared with that of the reconstructed incumbent. Of 589 transformer-zone clusters, 554 were electrically feasible. The value of reconfiguration was 0.01%, 0.39%, and 0.20% of like-for-like life-cycle cost in the three zones, all within a prespecified 1% practical-equivalence margin, and it stayed below 1% under seven alternative engineering assumptions. Value was positive in only 24 clusters, reaching at most 6.1% in a single cluster, and its apparent association with urban form reduced to

cluster size. Full alternating-current power-flow checks agreed with the linear feasibility screen for all 672 feeder–scenario cases. Under these reconstructed conditions, like-for-like renewal is a defensible default, and the method is best used to identify the small set of renewal sites where detailed reconfiguration engineering may be justified.

Practical Applications

When a city replaces overhead power lines with underground cables, the excavation needed for the new ducts also creates an opportunity to change how the local low-voltage network is laid out. Planners often assume that a redesigned layout should be cheaper or better. This study tests that assumption in three Japanese districts. Public data were used to approximate the existing network, engineering checks ensured that every alternative could carry the load within voltage and equipment limits, and life-cycle cost was used to compare alternatives. Keeping the existing layout was always allowed as an option. In most transformer zones, no feasible redesign was cheaper than replacing the existing layout like for like, and in each district the potential saving was below 0.4% of life-cycle cost. A few larger zones showed savings of several percent. The practical implication for utilities and municipalities is that renewal programs need not commission detailed redesign studies everywhere. A low-cost screening step can flag the few locations where redesign may pay, and these locations should then be checked with actual utility records and detailed engineering before any investment decision is made.

Author keywords: Infrastructure renewal; Asset management; Undergrounding; Distribution network reconfiguration; Life-cycle cost; Network reconstruction; Engineering feasibility.

Introduction

Much of the civil infrastructure that serves cities is approaching the end of its design life, and renewal programs for water, gas, transport, and electricity networks commit large public and

private budgets to replacing aging assets (Abu-Samra et al. 2020; Kerwin and Adey 2021). Renewal decisions are usually framed as asset-level questions of when and how to replace components (Hastak and Gokhale 2000; Khan et al. 2015) or as network-level scheduling problems that coordinate interventions across interdependent assets (Dao et al. 2019). A scheduled renewal also creates a less frequently examined opportunity: because excavation, duct installation, and reinstatement are paid for anyway, the renewal window is a rare moment at which the spatial configuration of a buried network can be changed at marginal rather than full cost.

The conversion of overhead electricity distribution to underground cable is a clear example. Economic studies show that blanket undergrounding rarely passes a cost–benefit test while targeted undergrounding can (Fenrick and Getachew 2012; Larsen 2016; Taheri et al. 2023). Long-horizon studies of power distribution reliability likewise treat undergrounding as an infrastructure investment strategy (Salman and Li 2017). Once a corridor has been committed to undergrounding, however, a narrower marginal question remains for the low-voltage network: should the inherited layout be replaced like for like, or should the committed civil works be used to re-lay the network along a different, possibly shorter or lower-loss, topology?

A technically feasible redesign is not necessarily an economically justified one. Distribution-planning methods have long optimized feeder routing, conductor selection, and reconfiguration (Tram and Wall 1988; Baran and Wu 1989; Samui et al. 2012), and recent infrastructure-systems work optimizes power-network investment under engineering constraints (Bynum et al. 2021). These formulations typically compare an optimized network with a greenfield design or with a forced alternative. When the comparison excludes the incumbent configuration, any redesign that is feasible will appear to offer some saving or some loss, and the

reported difference conflates genuine decision value with the arbitrary choice of comparator. The relevant renewal question is different: the incumbent layout remains an admissible decision, and a redesign has value only to the extent that it improves on that default.

This framing has a direct analogue in the infrastructure-systems literature on flexibility. Flexibility in network design has been valued as the cost difference between adaptable and committed plans (Basupi and Kapelan 2015), adaptive planning pathways have been proposed to avoid costly redesigns (Hadjidemetriou et al. 2022), and real-options methods have been used to value underground infrastructure investment under uncertainty (Zahed et al. 2020). The present study applies the same logic to a one-shot decision at renewal: the option to reconfigure is valued against the option to retain. Because the incumbent is always admissible, this value cannot be negative by construction; non-negativity is not an empirical finding. The empirical questions are how large the value is, how it is distributed across the network, and whether it is large enough relative to modeling uncertainty to justify detailed engineering.

A second difficulty is that the incumbent configuration is often poorly recorded. Infrastructure managers routinely make decisions with incomplete or missing asset data (Moghtadernejad et al. 2023; Dehghani et al. 2025), and the value of acquiring better data can itself be evaluated as a life-cycle cost decision (Leo and Swei 2021). For buried utility networks, synthetic or reconstructed models built from road networks and other proxies are increasingly used where utility records are unavailable (Bandam et al. 2021; Zhang et al. 2025; Liu et al. 2025; Babli et al. 2025), and their fidelity limits have been documented against ground truth (Fletcher et al. 2026). Utility networks are known to follow the morphology of the street network (Abdel-Mottaleb and Zhang 2020). Reconstruction therefore offers a way to screen renewal

decisions before detailed records are assembled, provided that its uncertainty is carried through to the decision and its conditionality is stated.

This paper develops and applies a feasibility-gated screening method for renewal-time reconfiguration of low-voltage distribution networks during undergrounding. The study asks three questions: (1) how large is the economic value of reconfiguration at renewal once feasibility and the incumbent option are respected; (2) where, if anywhere, is that value concentrated; and (3) how robust are these conclusions to engineering, economic, and electrical-model assumptions. The contribution is not a new optimization algorithm. It is a decision estimand and screening procedure for infrastructure renewal, together with evidence on the size and distribution of that value under reconstructed-network conditions (Fig. 1).

Methods

Study Design and Zones

Three Japanese municipalities were selected purposively for the availability of public spatial data relevant to reconstructing low-voltage network layouts, not to represent Japanese municipalities statistically (Table 1). Sano (Tochigi Prefecture) publishes security-light records with pole numbers and coordinates (observability tier 1; Sano City n.d.); Kudamatsu (Yamaguchi Prefecture) publishes geolocated security-light records without verified pole linkage (tier 2; Kudamatsu City n.d.); and Yasu (Shiga Prefecture) is represented by road and building geometry only (tier 3). In each municipality a compact study zone was extracted as the largest connected component of the OpenStreetMap road graph (OpenStreetMap Contributors 2026) (Fig. 2). Building footprints were assigned an estimated peak demand of 30–60 W/m² over two floors and snapped to the nearest road node; mean snap distances of 37–49 m are treated as error in the modeled demand surface. Buildings whose raw peak exceeded 150.0 kW were treated as served

at medium voltage and excluded from low-voltage synthesis. Demand nodes were grouped into transformer-zone clusters, which are the unit of decision and analysis.

Reconstruction of Incumbent Networks

Because actual low-voltage cable routes are unobserved, each cluster's incumbent network was reconstructed as an ensemble of radial trees on the road graph, which serves as the admissible conduit space. Security-light anchors, where available, were used as routing priors on the demand side and were never equated with pole or cable locations; the evidence available in each tier determined how strongly the reconstruction was constrained. The primary ensemble comprises 12 replicates per cluster drawn from two generator families; a 24-replicate ensemble and leave-one-family-out variants were used to check stability. The incumbent was evaluated with the 150 mm² reference conductor throughout, representing like-for-like replacement of the inherited configuration.

Candidate Redesigns

Redesign candidates were generated for each cluster as weighted Steiner trees connecting the transformer source and all demand terminals on the local road graph, using the Kou–Markowsky–Berman heuristic (Kou et al. 1981). Edge weights combined route length with a load-dependent loss penalty, and six penalty weights (0, 1, 3, 10, 30, and 100) produced a family of candidates ranging from minimum-length to loss-averse trees. Each candidate tree was pruned to its terminals, and conductor sizes were selected edge by edge from 150, 250, 400, 600 mm² to minimize capital plus discounted loss cost subject to ampacity. The best redesign is therefore the least-cost member of an enumerated candidate set, not a proven optimum over all possible layouts; the resulting value is a lower bound on what an exhaustive search could find under the same model.

Engineering Feasibility Gates

Every incumbent replicate and every redesign candidate was evaluated with the same radial branch-flow calculation at 415 V line-to-line, power factor 0.9, and aluminium resistivity 2.826×10^{-8} Ω·m. Demand was diversified with a coincidence factor of 0.6. An alternative was feasible only if it was radial and connected, no branch current exceeded the ampacity of its conductor (350 A for 150 mm², 460 A for 250 mm², 600 A for 400 mm², 780 A for 600 mm²), the coincident cluster load did not exceed the 150 kVA transformer rating, and the cumulative voltage drop from the source to any node did not exceed 6%. A cluster entered the economic comparison only if its least-cost redesign was feasible and at least one incumbent replicate passed the ampacity gate with the reference conductor; because the transformer load is common to both alternatives and the incumbent voltage drops in feasible clusters were at most 4.93%, every incumbent replicate in the comparison set satisfied all gates. If the least-cost redesign violated the voltage gate, the cluster was excluded even when a more expensive candidate might have been feasible; this rule is conservative with respect to the value of reconfiguration.

Life-Cycle Cost Model

The net present cost (NPC) of an alternative S was computed as

$$NPC(S) = C(S) + \sum_{t=1}^{H} \frac{p_e E(S) + \omega C(S)}{(1+i)^t} \qquad (1)$$

where C(S) is the capital cost of trenching, duct bank, and cable plus the transformer unit, E(S) is the annual energy loss, p_e is the value of energy, ω is the annual operation and maintenance fraction of capital cost, i is the discount rate, and H is the horizon. The frozen primary values are H = 40 years, i = 3%, p_e = 25 JPY/kWh, ω = 1.5%, excavation 120,000 JPY/m, duct bank 30,000 JPY/m, reference cable 12,000 JPY/m (scaled by conductor area), and

transformer 3,500,000 JPY. Annual losses were computed from peak branch losses with a load factor of 0.35. All costs are in Japanese yen at constant prices.

Value of Reconfiguration

For cluster c with incumbent replicates r = 1, …, R_c and redesign candidates k ∈ K_c, the value of reconfiguration (VoF) is

$$VoF_c = \frac{1}{R_c} \sum_{r=1}^{R_c} max\{0, NPC(S1_{c,r}) - \min_{k \in K_c} NPC(S2_{c,k})\} \qquad (2)$$

Retaining the incumbent is an admissible decision, so VoF_c is zero whenever no feasible redesign is cheaper than the reconstructed incumbent and positive otherwise. Averaging over replicates carries reconstruction uncertainty into the estimand. Because the redesign may change both topology and conductor sizes, VoF measures the value of re-layout with re-sizing; a topology-only variant with the reference conductor was computed as an ablation. Zone-level VoF is reported as a cost-weighted share of like-for-like NPC,

$$VoF\%_z = 100 \frac{\sum_{c \in F_z} VoF_c}{\sum_{c \in F_z} \overline{NPC}(S1_c)} \qquad (3)$$

where F_z is the set of feasible clusters in zone z and the bar denotes the replicate mean. Cluster VoF values below 10^{-6}% of incumbent NPC were treated as numerical zeros. For comparison, a secondary naive contrast was computed as the difference between mean incumbent NPC and the NPC of the best redesign without the max operator, that is, with the incumbent excluded from the choice set; this contrast can be negative and is reported only to show the effect of excluding the incumbent.

Equivalence Margin and Sensitivity Analyses

Before the results were examined, a practical-equivalence margin of 1% of like-for-like NPC was fixed: zone-level reconfiguration value below this margin is interpreted as economically equivalent to like-for-like renewal, values above it as beneficial, and estimates whose uncertainty crosses the margin as indeterminate. Engineering-assumption sensitivity was assessed by re-running the full pipeline with the medium-voltage threshold at 120 and 200 kW, the cluster target load at 80 kW, the coincidence factor at 0.5, 0.7, and 0.8, and the 24-replicate ensemble. Boundary variants with 80 and 130 kVA transformer limits were also examined. Economic sensitivity of the naive contrast was assessed by a one-at-a-time tornado analysis over discount rate, energy value, horizon, load factor, operation and maintenance fraction, and cost multiplier, and by a 500-draw Monte Carlo analysis.

Alternating-Current Validation

The linear branch-flow model (Gan and Low 2014) was compared with full alternating-current (AC) power flow in pandapower (Thurner et al. 2018) for every feasible cluster with a genuine topology choice. For each such cluster, the first incumbent replicate and a least-cost redesign were modeled with a slack bus at the source, coincident loads at the stated power factor, per-edge resistance for the reference conductor, and a reactance of 0.08 Ω/km. Agreement was assessed by the voltage-magnitude difference, the relative loss error, and agreement on the 6% voltage-drop gate. The primary check loads each feeder only at its own cluster terminals, as in the economic comparison. A stress case that also loads customer nodes of neighboring clusters lying on the reconstructed incumbent tree is reported separately.

Conditions Associated with Positive Value

To indicate where detailed engineering might be worthwhile, cluster VoF was related descriptively to ten street-network and demand descriptors (Cardillo et al. 2006; Boeing 2020), including terminal count, coincident demand, convex-hull area, demand density on hull and 100-m buffer supports, road density, and incumbent route length per terminal. Pooled Spearman rank correlations were computed with and without rank-residual adjustment for terminal count. Spatial dependence of cluster VoF within each zone was summarized by Moran's I with inverse-distance weights and 200 permutations. Because clusters are nested in three purposive zones and the distribution of VoF is dominated by zeros, these analyses are descriptive.

Results

Reconstructed Infrastructure and Feasibility

The three study zones contain 589 transformer-zone clusters with 58.6 MW of modeled low-voltage peak demand (Table 1). Of these, 554 clusters (49.6 MW) passed the feasibility gates and entered the economic comparison: 360 in Sano, 88 in Kudamatsu, and 106 in Yasu (Table 2). 218 of the feasible clusters serve a single customer node and therefore offer no topology choice; the remaining 336 clusters have a genuine choice between the incumbent and at least one alternative layout. Across feasible clusters, the least-cost redesign seldom changed the corridor length materially relative to the reconstructed incumbent (Fig. S1 in the Supplemental Materials).

Magnitude and Distribution of Reconfiguration Value

Zone-level VoF was 0.01%, 0.39%, and 0.20% of like-for-like NPC in Sano, Kudamatsu, and Yasu, respectively (Table 2). All three values lie well inside the 1% practical-equivalence

margin (Fig. 3). Restricting the denominator to clusters with a topology choice changes these values only slightly (0.02%, 0.40%, and 0.21%). VoF was positive in 24 of 554 feasible clusters (4.3%), or 7.1% of the clusters with a topology choice (Table 3; Fig. 4). Conditional on being positive, the median cluster VoF was 0.49%, 0.58%, and 0.24% of cluster NPC, and the maximum was 1.07%, 5.08%, and 6.10%.

The secondary naive contrast, which excludes the incumbent from the choice set, was 0.01%, -10.06%, and 0.20% (Table 2). The large negative Kudamatsu value arises because a few small, spatially extended clusters have no redesign cheaper than the incumbent corridor; forcing a redesign there adds cost without compensating loss reduction. This value is an artifact of the comparator, not evidence that reconfiguration is harmful when the incumbent remains available.

Geographic Variation

VoF was lowest in Sano, the zone with the most constrained reconstruction (tier 1), and highest in Kudamatsu (tier 2), with Yasu (tier 3) intermediate (Fig. 3). The positive-value incidence followed the same order: 3 of 360 feasible clusters in Sano, 13 of 88 in Kudamatsu, and 8 of 106 in Yasu (Table 3). Because observability tier, settlement form, and demand density vary together across only three zones, these differences cannot be attributed to any single factor; in particular, weaker observability may admit more reconstructed incumbents that a redesign can improve on. Within zones, Moran's I of cluster VoF ranged from -0.023 to -0.001 (smallest permutation $p = 0.15$), giving no indication of spatial clustering, although a non-significant result in zero-dominated data cannot prove its absence.

Conditions Associated with Positive Value

In pooled rank correlations, cluster VoF increased with the number of demand terminals ($\rho = 0.32$), spatial span (0.29), convex-hull area (0.29), and incumbent route length per terminal

(0.27) (Table 4). After rank-residual adjustment for terminal count, these associations fell to -0.01, 0.01, and -0.03, respectively. Demand density was not consistent across spatial supports (hull $\rho = -0.28$; buffer $\rho = -0.04$). Larger clusters mechanically offer more alternative routes, and no independent urban-form correlate of reconfiguration value was found beyond cluster size.

Robustness and Sensitivity

Across the 7 engineering-assumption variants, zone VoF ranged from 0.001% to 0.947% and never reached the 1% margin (Table 5). The largest values occurred with a 200 kW medium-voltage threshold and the smallest with an 80 kW cluster target. The maximum voltage drop among feasible clusters in any variant was 5.91%. The 24-replicate ensemble changed zone VoF by at most 0.040 percentage points. The economic sensitivity analyses apply to the naive contrast: in Kudamatsu all 500 Monte Carlo draws remained negative (5th–95th percentile -0.58 to -0.31 billion JPY), whereas in Sano and Yasu the contrast remained positive but small, so the sign pattern is not an artifact of a particular price or discount assumption. Decomposition of the incumbent-minus-redesign difference into capital, discounted loss, and residual terms shows that capital differences dominate and loss savings are small at low-voltage scale (Fig. 5).

Engineering Validation

Full AC power flow converged for all 672 feeder–scenario objects in 336 clusters (Table 6). With feeders loaded at their own cluster terminals, the linear screen and AC power flow agreed on feasibility in every case (672 of 672 feasible under both); the maximum voltage-magnitude difference was 0.0056 per unit, the maximum relative loss error was 0.079, and the maximum voltage drop was 4.27% (linear) and 4.82% (AC), leaving at least 1.18 percentage points of margin to the 6% gate. In the stress case, feasibility agreement remained complete, but two incumbent replicates in Sano exceeded the gate under both models (maximum drop 9.43%

linear and 11.36% AC), and the maximum voltage difference rose to 0.0193 per unit. These cases illustrate how sensitive voltage feasibility is to the reconstructed load assignment, but they do not enter the economic comparison.

Discussion

Why Feasible Redesign Yields Little Economic Benefit

The central result is that, once engineering feasibility and the incumbent option are respected, renewal-time reconfiguration of these low-voltage networks offers little economic value. First, life-cycle cost at low voltage is dominated by corridor length through excavation and duct costs, while loss costs are small; a redesign must therefore shorten the corridor to pay, and on a road-constrained conduit space the incumbent route is usually already close to the shortest feasible connection. Second, the feasibility gates remove many apparently attractive alternatives: treating raw building peaks as undiversified loads or ignoring voltage limits creates fictitious headroom and therefore fictitious savings.

Incumbent-Admissible Comparison

The difference between the primary estimand and the naive contrast is substantive for infrastructure decision-making. In Kudamatsu, a forced-redesign comparison reports a cost penalty of about 10%, whereas the incumbent-admissible estimand shows that the zone contains several clusters where redesign is worthwhile and many where it is simply not exercised. This mirrors the treatment of flexibility as an option in infrastructure planning (Basupi and Kapelan 2015; Zahed et al. 2020). The contribution of the present study is not the non-negativity of VoF, which follows from the formulation, but the demonstration that its magnitude can be estimated

from reconstructed networks under explicit feasibility gates and compared with a prespecified equivalence margin.

Use as a Screening Step under Incomplete Records

The method is intended as a screening step in renewal programs, not as a design tool. Its inputs are open data, its outputs identify the clusters where detailed reconfiguration engineering might be worth commissioning, and its zero-dominated results imply that most clusters can proceed with like-for-like renewal. As with data procurement decisions (Leo and Swei 2021), screening is useful if the cost of detailed engineering in the flagged clusters is smaller than the value it uncovers. The present study did not measure the cost of operating such a screening program or the savings that a utility would realize; the screening value is therefore a hypothesis supported by the concentration of VoF, not a measured benefit. Integration with existing renewal scheduling and coordinated work packaging (Dao et al. 2019; Abu-Samra et al. 2020; Kerwin and Adey 2021) would allow flagged clusters to be prioritized when corridor works are already planned.

Evidence Needed before Operational Deployment

Before the method could support utility investment decisions, reconstructed incumbents should be compared with utility network records for a sample of clusters, as has been done for synthetic water distribution models (Fletcher et al. 2026); unit costs and construction constraints such as traffic management, other buried utilities, and customer outages should be calibrated with utilities; flagged clusters should be checked by detailed design; and a larger, less purposive set of zones should be analyzed to test whether the low magnitudes found here are typical.

Limitations

The results are conditional on reconstructed rather than fully observed networks: security-light anchors and road graphs do not observe actual cable routes, and reconstruction uncertainty is represented only by the replicate ensemble. The three zones were selected purposively for data availability, differ in observability, and do not support inference to Japanese or other municipalities. Demand, coincidence, the voltage-drop limit, the low-voltage/medium-voltage boundary, and unit costs are modeling assumptions; several of them move the naive contrast materially, although none moved zone VoF above the margin. Economic uncertainty intervals were computed for the naive contrast rather than for VoF itself. The electrical model is a balanced three-phase radial approximation without unbalance, harmonics, protection coordination, or future load growth and distributed generation. The candidate set is enumerated rather than exhaustive. The method has not been validated against utility investment decisions, and institutional, regulatory, and construction constraints that would restrict or enable reconfiguration in practice were not modeled. The magnitudes reported here should therefore be read as modeled, zone-specific screening estimates rather than realized savings.

Conclusions

Scheduled renewal creates an opportunity to reconfigure buried infrastructure at marginal cost, but a feasible redesign is not automatically a worthwhile one. This study defined the value of reconfiguration at renewal as the expected life-cycle cost saving of the best feasible redesign relative to an admissible incumbent, and estimated it for reconstructed low-voltage distribution networks in three Japanese undergrounding study zones. Zone-level value was 0.01%, 0.39%, and 0.20% of like-for-like life-cycle cost, within a prespecified 1% equivalence margin and robust to engineering and economic assumptions; positive value occurred in 4.3% of feasible

clusters and was concentrated in larger clusters. For infrastructure managers, the implication is that like-for-like renewal is a defensible default for such networks and that reconfiguration assessment should be targeted rather than universal. The broader contribution is a feasibility-gated, incumbent-admissible screening procedure that can be applied with open data where records are incomplete.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository online in accordance with funder data retention policies.

Some or all data, models, or code used during the study were provided by a third party. Direct requests for these materials may be made to the provider as indicated in the Acknowledgements.

The analysis code, configuration files, source registry with checksums, and all generated machine-readable results (cluster-level results, sensitivity analyses, AC validation tables, and morphology and spatial diagnostics) are available in a public repository (Onishi 2026). Road and building data are from OpenStreetMap under the Open Database License (OpenStreetMap Contributors 2026); security-light records are municipal open data from Kudamatsu City under CC BY (Kudamatsu City n.d.) and from Sano City under its open-data terms (Sano City n.d.); map layers use geoBoundaries (CC BY 4.0; Runfola et al. 2020) and Natural Earth (public domain; Natural Earth n.d.). Third-party raw inputs are not redistributed where their terms do not permit it; their source URLs, retrieval dates, and checksums are documented in the repository so that they can be retrieved from the providers.

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Disclosure of AI Tool Use

During the preparation of this work, the author used an AI-based coding and writing assistant (Devin, Cognition AI) to scaffold and review analysis code, check numerical consistency between outputs and text, retrieve and verify bibliographic metadata, and draft and edit portions of the manuscript text. The author reviewed and edited all content, verified all reported results against the generated outputs, and takes full responsibility for the content of the publication. No AI tool is listed as an author.

Supplemental Materials

Fig. S1 is available online in the ASCE Library (www.ascelibrary.org).

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Figure Captions

Fig. 1. Data-to-decision pipeline from open geospatial inputs to the incumbent-admissible renewal comparison.

Fig. 2. Study zones: (a) locations in Japan (boundaries: geoBoundaries, CC BY 4.0; coastlines: Natural Earth); (b–d) road graph and snapped demand nodes in Sano, Kudamatsu, and Yasu (map data © OpenStreetMap contributors, ODbL).

Fig. 3. Value of reconfiguration by zone (% of like-for-like NPC) and Monte Carlo mean of the naive contrast; shaded band: ±1% equivalence margin.

Fig. 4. Cluster-level value of reconfiguration: positive-value incidence by zone and magnitudes for the 24 clusters with positive value.

Fig. 5. Zone-level incumbent-minus-redesign cost difference by component (billion JPY, 40-year horizon) and aggregate value of reconfiguration; panels use independent vertical scales.

Table 1. Study zones, observability tiers, and cluster counts

Zone	Observability tier	LV demand, all clusters (MW)	MV-excluded demand (MW)	Clusters	Feasible clusters	Feasible single-customer clusters
Sano	1: pole-linked anchors	38.4	12.2	385	360	175
Kudamatsu	2: geolocated anchors	9.5	7.2	96	88	19
Yasu	3: roads and buildings only	10.7	8.9	108	106	24
Total		58.6		589	554	218

Note: LV = low voltage; MV = medium voltage. MV-excluded demand is from buildings above the LV/MV threshold.

Table 2. Zone-level life-cycle cost and value of reconfiguration (feasible clusters)

Zone	Feasible clusters	Feasible demand (MW)	NPC like-for-like (billion JPY)	NPC best redesign (billion JPY)	VoF (million JPY)	VoF (% of like-for-like NPC)	VoF, clusters with topology choice (%)	Naive contrast (%)
Sano	360	31.8	4.84	4.84	0.7	0.014	0.017	0.01
Kudamatsu	88	7.6	4.65	5.12	18.3	0.394	0.401	−10.06
Yasu	106	10.2	3.00	2.99	6.0	0.201	0.209	0.20

Note: NPC = net present cost, 40-year horizon, 3% discount rate. The naive contrast excludes the incumbent from the choice set and is reported for comparison only.

Table 3. Distribution of cluster-level value of reconfiguration

Zone	Feasible clusters	Single-customer (no choice)	With topology choice	Positive VoF	Zero share, all feasible	Zero share, with choice	Median VoF if positive (%)	Maximum VoF (%)
Sano	360	175	185	3	0.992	0.984	0.486	1.066
Kudamatsu	88	19	69	13	0.852	0.812	0.582	5.079
Yasu	106	24	82	8	0.925	0.902	0.241	6.098
Total	554	218	336	24	0.957	0.929		

Note: Cluster VoF below 0.000001% of incumbent NPC is treated as zero.

Table 4. Rank correlation of cluster value of reconfiguration with network and demand descriptors

Descriptor	Pooled Spearman ρ	ρ adjusted for terminal count	n
Demand terminals (count)	0.32	—	554
Coincident demand (kW)	0.13	−0.03	554
Convex-hull area (km²)	0.29	0.01	127
Demand density, hull (kW/km²)	−0.28	−0.04	127
Demand density, 100-m buffer (kW/km²)	−0.04	−0.06	554

Road density (km/km²)	0.23	−0.19	554
Mean road-edge length (m)	0.16	0.08	296
Mean road-node degree	0.29	−0.29	296
Incumbent route length per terminal (m)	0.27	−0.03	554
Spatial span (m)	0.29	−0.01	554
Topology divergence (endogenous diagnostic)	0.37	0.78	554

Note: Descriptive only; clusters are spatially dependent and nested in three zones. n below the feasible total reflects degenerate hulls or clusters without internal road edges. Topology divergence is derived from the reconstruction itself and is not an urban-form predictor.

Table 5. Sensitivity of zone-level value of reconfiguration to engineering assumptions

Variant	**Sano feasible**	**Sano VoF (%)**	**Kudamatsu feasible**	**Kudamatsu VoF (%)**	**Yasu feasible**	**Yasu VoF (%)**
Baseline	360	0.014	88	0.394	106	0.201
MV threshold 120 kW	307	0.094	78	0.201	100	0.782
MV threshold 200 kW	405	0.074	94	0.561	121	0.947
Cluster target 80 kW	472	0.001	113	0.441	131	0.107
Coincidence factor 0.5	377	0.073	93	0.486	108	0.196
Coincidence factor 0.7	350	0.016	82	0.319	97	0.226
Coincidence factor 0.8	327	0.019	78	0.313	88	0.063
24-replicate ensemble	360	0.003	88	0.380	106	0.241

Note: Each variant re-runs reconstruction, gating, and optimization. Prespecified equivalence margin: 1%.

Table 6. Linear feasibility screen versus full AC power flow

Statistic	**Cluster-terminal loading (primary)**	**All-node loading (stress case)**
Feeder–scenario objects (clusters)	672 (336)	672 (336)
Non-converged	0	0
Feasibility agreement	1.000	1.000
Feasible under both models	672	670
Median $\|\Delta V\|$ (per unit)	0.0001	0.0001
Maximum $\|\Delta V\|$ (per unit)	0.0056	0.0193
Maximum relative loss error	0.079	0.226
Maximum voltage drop, linear (%)	4.27	9.43
Maximum voltage drop, AC (%)	4.82	11.36

Note: Cluster-terminal loading matches the economic comparison; the stress case also loads customer nodes of neighboring clusters on the reconstructed incumbent tree. Voltage-drop gate: 6%.